

Acute Viral Hepatitis

ILOs

By the end of lecture, the students should be aware about:

- Definition of acute viral hepatitis and epidemiology
- How do you test for and diagnose acute hepatitis?
- How to prevent and which vaccine is given for viral hepatitis?
- What is the best treatment for viral hepatitis?

Definition: Acute hepatitis is a term used to describe a wide variety of conditions characterized by acute inflammation of the hepatic parenchyma or injury to hepatocytes resulting in elevated liver function indices.

In general, hepatitis is classified as acute or chronic based on the duration of the inflammation and insult to the hepatic parenchyma. If the period of inflammation or hepatocellular injury lasts for less than six months, characterized by normalization of the liver function tests, it is called acute hepatitis.

Epidemiology

Generally, the rates of viral hepatitis are low in high-income regions and high in resource-poor areas. The past five years have seen a small increase in cases mostly from isolated food-related outbreaks, persons who use drugs, and the homeless population. In high endemic areas, HAV has affected more than 90% of children by age 10.

Causes of Acute Viral Hepatitis

Acute viral hepatitis can be caused by five major hepatotropic hepatitis viruses(A, B, C, D, E):

Viral Hepatitis Over View				
Virus	Source	Transmission	Chronicity	Prevention
A	Stool	Feco-oral	No	Immunoglobulin Vaccination
B	Parenteral	Percutaneous per mucosal	Yes	Immunoglobulin Vaccination
C	Parenteral	Percutaneous per mucosal	Yes	Blood donor screening Risk behavior modification
D	Parenteral	Percutaneous per mucosal	Yes	HBV Prevention
E	Stool	Feco-oral	No	Safe water cleaning

Symptoms of Acute Viral Hepatitis

- Acute viral hepatitis can cause anything from a minor flu-like illness to fatal liver failure. Sometimes there are no symptoms. The severity of symptoms and speed of recovery vary considerably, depending on the virus and the person's response to the infection.
- Hepatitis A and C often cause very mild symptoms or none at all and may be unnoticed. Hepatitis B and E are more likely to produce severe symptoms. Symptoms of acute viral hepatitis usually begin suddenly. They include a poor appetite, malaise, nausea and vomiting, fever...etc.
- Sometimes, after 3 to 10 days, the urine becomes dark, and stool becomes pale. Jaundice (a yellowish discoloration of the skin and whites of the eyes) may develop. It is sometimes accompanied by itching. The liver may be enlarged and tender.
- Most of the initial symptoms usually disappear within a week, and people feel better even though the jaundice may worsen.
- Rarely, **fulminant hepatitis** develops. Fulminant hepatitis is severe hepatitis with signs of liver failure. Fulminant hepatitis can occur in people with hepatitis A, but it is more likely to develop in people with hepatitis B, particularly if they also have hepatitis D.
- People with acute viral hepatitis usually recover in 4 to 8 weeks, even without treatment. However, some people infected with hepatitis B or C develop chronic infections.

Diagnosis of Acute Viral Hepatitis

Blood tests

Liver tests involving measuring the levels of liver enzymes and other substances produced by the liver. These blood tests can detect parts of specific viruses or specific antibodies produced by the body to fight the viruses. Specific laboratory tests include:

HAV:

- HAV: IgM anti-HAV: positive at the time of onset of symptoms; results remain positive for 3-6 months after the primary infection
- Anti-HAV IgG appears soon after IgM and generally persists for many years.

HBV:

- Serology: HBsAg, HBeAg, HBeAb, HBcAb, HBsAb, HBcAb, HBsAb, HBcAb
- HBV Viral DNA: Most accurate marker of viral DNA and detected by PCR

HCV:

- HCV: Anti-HCV; cannot distinguish acute from chronic infection

- PCR: used to detect viral RNA → HCV (the most accurate)

Hepatitis E:

- Anti-HEV antibodies: IgM (acute), IgG (chronic)
- Polymerase chain detection assay (PCR)

Prevention of Acute Viral Hepatitis

Because treatments for viral hepatitis are only partially effective, preventing viral hepatitis is very important. Prevention of Acute Viral Hepatitis include: **Vaccines and immune globulin**

1-Vaccines

- Vaccines to prevent hepatitis A and hepatitis B are available. A vaccine for hepatitis E is currently available only in China. No vaccines against hepatitis C or D virus are available. However, vaccination against hepatitis B virus also reduces the risk of infection with hepatitis D virus. Hepatitis vaccines are given by injection into muscle.
- Routine vaccination with the hepatitis A vaccine and hepatitis B vaccine is recommended for all children and for adults at high risk of getting hepatitis.
- As with most vaccines, protection requires allowing several weeks for the vaccine to reach its full effect as the immune system gradually creates antibodies against the virus.

2-Immune globulin

If people who have not been vaccinated are exposed to hepatitis B virus, they are given hepatitis B immune globulin and are vaccinated. It is given by injection. Babies born to mothers with hepatitis B are given hepatitis B immune globulin (injected into a muscle) and hepatitis B vaccine.

Other preventive measures against infection with the hepatitis viruses can be taken:

- Washing hands thoroughly before handling food
- Not sharing needles to inject drugs
- Not sharing toothbrushes, razors, or other items that could get blood on them
- Practicing safe sex—for example, using barrier protection such as a condom, Limiting the number of sex partners

Donated blood is unlikely to be contaminated because it is screened. Nonetheless, doctors help to reduce the risk of hepatitis by ordering blood transfusions only when essential. Before surgery, people can also sometimes prevent the need for transfusion of blood from an unknown donor by donating their own blood weeks before the operation.

Treatment of Acute Viral Hepatitis

- **Supportive care**

For most people with acute viral hepatitis, special treatment is not necessary. However, people with severe acute hepatitis may require hospitalization so that symptoms can be treated. If doctors suspect that fulminant hepatitis is developing, the person is hospitalized so that mental status can be monitored, liver tests can be done, and doctors can determine whether liver transplantation is needed.

After the first several days, appetite usually returns, and people do not need to stay in bed. Severe restrictions of diet or activity are unnecessary, and vitamin supplements are not required. Most people can safely return to work after the jaundice clears, even if their liver test results are not quite normal.

People with hepatitis should not drink alcohol until they have fully recovered.

The infected liver may not process (metabolize) drugs normally. So a doctor may need to stop a drug or reduce the dosage of a drug that could accumulate to harmful levels in the body (such as warfarin or theophylline). Thus, people with hepatitis should tell their doctor all the drugs they are taking (both prescription and nonprescription, including any medicinal herbs), so that the dosage of the drug can be adjusted if necessary.

If itching occurs, cholestyramine, taken by mouth, is often effective.

If hepatitis B causes fulminant hepatitis, people are usually treated in an intensive care unit. Antiviral drugs may help. However, liver transplantation is the most effective treatment and is the best hope of survival, particularly for adults.

Questions:

What is definition of acute viral hepatitis?

How do you test for and diagnose acute hepatitis?

How to prevent and which vaccine is given for viral hepatitis?

What is the treatment for viral hepatitis?

What are the sequelae of acute viral hepatitis?